

DHCP Lab

TITLE : DHCP Configuration in Cisco Packet Tracer

Aim : To configure a router as a DHCP server so that PCs can automatically obtain IP addresses.

NETWORK DEVICES USED :

1. Router – 1
2. Switch – 1
3. PC – 4 (PC0, PC1, PC2, PC3)
4. Data Cables – Copper Straight Through

NETWORK ADDRESSING :

- Network Address : 192.168.10.0
- Subnet Mask : 255.255.255.0
- Router Interface (Connected to Switch) : 192.168.10.1

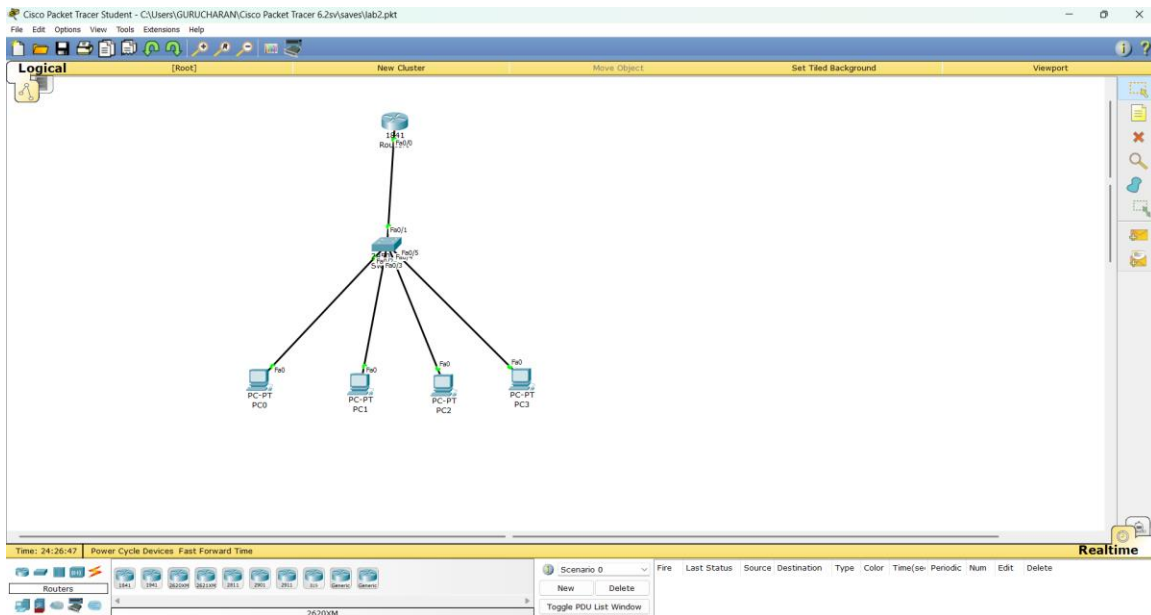
PROCEDURE :

Step 1 : Create Network Topology

1. Open Cisco Packet Tracer.
2. Add 1 Router, 1 Switch and 4 PCs.
3. Connect all devices using Copper Straight Through cables.

Connections :

- PC0 → Switch
- PC1 → Switch
- PC2 → Switch
- PC3 → Switch
- Router → Switch



Step 2 : Configure Router IP Address

- Open Router CLI and enter the following commands :
- enable
- configure terminal
- interface fa0/0
- ip address 192.168.10.1 255.255.255.0
- no shutdown
- exit

```

c:\network
Physical  Config  CLI

IOS Command Line Interface

Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
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Compiled Wed 18-Jul-07 04:52 by pt_team
Image text-base: 0x60080600, data-base: 0x6270C200

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http://www.cisco.com/wu/export/crypto/tool/stqsg.html
If you require further assistance please contact us by sending email to
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Cisco 1841 (revision 3.0) with 114688K/16384K bytes of memory.
Processor board ID FT09471J02
M540 processor: part number 0, mask 49
3 HardEthernet/IEEE 802.3 interface(s)
1536K bytes of NVRAM.
43488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE SOFTWARE (fc2)
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Press RETURN to get started!

ALINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1#enable
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#int fa0/0
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#exit
R1#
M575-5-CONFIG_I: Configured from console by console
Copy Paste

```

Step 3 : Configure Router as DHCP Server

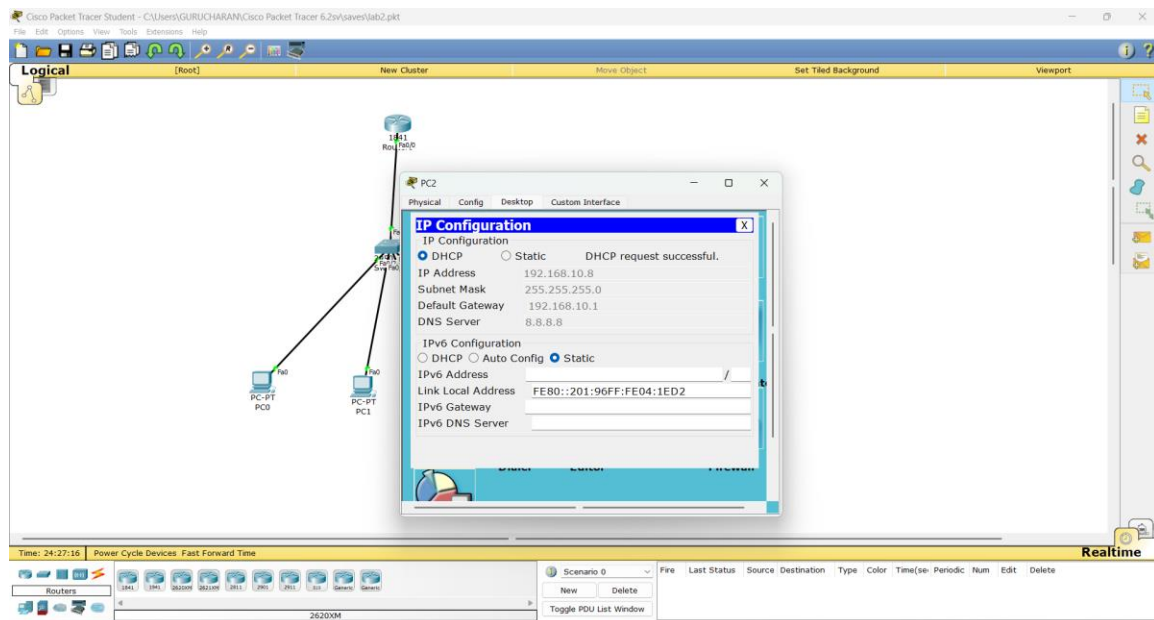
- configure terminal
- ip dhcp excluded-address 192.168.10.1 192.168.10.5
- ip dhcp pool LAN
- network 192.168.10.0 255.255.255.0
- default-router 192.168.10.1
- exit



Step 4 : Configure PCs to Obtain IP Automatically Open each PC (PC0, PC1, PC2, PC3).

- Go to Desktop → IP Configuration.
- Select DHCP option.

The PC will automatically receive an IP address from the router.



Step 5 : Verification

- Check IP address assigned to each PC.
- Expected range : 192.168.10.6 – 192.168.10.254.

Step 6 : Ping Test

- Open Command Prompt on any PC.
- Type ping followed by another PC IP address.

Example : ping 192.168.10.6

If replies are received, the network configuration is successful.

EXPECTED RESULT :

All PCs should automatically receive IP addresses from the DHCP server in the range 92.168.10.6 – 192.168.10.254 and should be able to communicate with each other.

